

PROJECT:  
*Small Multimodal  
Models for  
Clinical Diagnosis*

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# Module Info



Type of module

Project



Credit points

10



Weekly hours

4

# Objectives



Work on practical research problems



Expand knowledge in *Healthcare, Medicine, Diagnostics, ...*



Deepen knowledge in *Computer Science, Artificial Intelligence, Vision Language Models, ...*



Sharpen skills in *Software Engineering, Project Management, Teamwork, ...*



Strengthen interdisciplinary competencies in *Critical Thinking, Problem Solving, Decision Making, ...*

# Content

|                                                                                                        |                                         |
|--------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <i>Small<br/>Multimodal<br/>Models for<br/>Tropical &amp;<br/>Infectious<br/>Disease<br/>Diagnosis</i> | Tropical & Infectious Disease Diagnosis |
|                                                                                                        | Vision Language Models                  |
|                                                                                                        | Multimodal Data                         |
|                                                                                                        | Retrieval-Augmented Generation          |
|                                                                                                        | Knowledge Graph                         |
|                                                                                                        | Finetuning                              |
|                                                                                                        | Multi-Agents                            |

# Deliverables



## Diagnosis Support System for Tropical and Infectious Diseases using Multimodal Data

Clinical Diagnosis (Tropical and Infectious Cases)  
Multimodal Data (Language + Vision)  
SMMs (Small Multimodal Models)  
Multi-Agent Systems (MedAgents)



## Project Management Documentation

Agile Software Development  
DevOps  
AI Engineering



## Technical Report

Usecases  
Data  
Models

# Format

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Teamwork

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Class interaction

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Weekly demo

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Final presentation

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Project report

# Grading



Continuous

30%



Final

70%

# Hints

1. Data collection and processing
  - Text
  - Image
2. Methods in consideration
  - RAG/KG
  - Finetuning
  - Multi-Agents
3. Model choices
  - CLIP/BLIP
  - Gemma/LLaVA
4. Optimization techniques
  - PEFT, LoRA
  - Quantization, QLoRA
5. Evaluation frameworks
  - RAG/KG
  - Finetuning
  - Multi-Agent

# References

## Survey article

- Large Language Models for Disease Diagnosis: A Scoping Review

## Research article

- LightRAG: Simple and Fast Retrieval-Augmented Generation
- QLoRA: Efficient finetuning of quantized LLMs
- MedAgents: Large Language Models as Collaborators for Zero-shot Medical Reasoning

## Data from PubMed

- MultiCaRe - A Multimodal Clinical Case Dataset

## Models from Huggingface

- <https://huggingface.co>

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# Contacts